

PRAKTIKUM BASIS DATA

13_NoSQL – Document_Database



Oleh:

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JURNAL MODUL

Disediakan dataset berikut:

```
db.toko.insertMany([
  {
    "name" : "Toko Berkah",
    "category" : "Elektronik",
    "city" : "Jakarta",
    "address" : { "street" : "Jl. Sudirman", "zipcode" : "10220" },
    "products" : [
      { "item" : "TV", "price" : 3500000, "stock" : 10 },
      { "item" : "Kulkas", "price" : 2800000, "stock" : 5 }
    ]
  },
  {
    "name" : "Toko Maju",
    "category" : "Elektronik",
    "city" : "Bandung",
    "address" : { "street" : "Jl. Asia Afrika", "zipcode" : "40111" },
    "products" : [
      { "item" : "HP", "price" : 1500000, "stock" : 20 },
      { "item" : "Laptop", "price" : 8000000, "stock" : 3 }
    ]
  },
  {
    "name" : "Toko Salim Bakery",
    "category" : "Elektronik",
    "city" : "Jakarta",
    "address" : { "street" : "Jl. Sudirman", "zipcode" : "10220" },
    "products" : [
      { "item" : "Monitor LED", "price" : 7700000, "stock" : 11 },
      { "item" : "Airfryer Panasonic", "price" : 4850000, "stock" : 7 }
    ]
  },
  {
    "name" : "Toko Sudinotech",
    "category" : "Elektronik",
    "city" : "Surabaya",
    "address" : { "street" : "Jl. Sudirman", "zipcode" : "10220" },
    "products" : [
```

```

    { "item": "RTX 55000", "price": 8500000, "stock": 20 },
    { "item": "Set PC Gaming Radeon", "price": 12000000, "stock": 9 }
  ]
},
})

```

1.) Tambahkan 1 toko baru bebas ke koleksi toko, dengan ketentuan:

- Minimal memiliki 2 produk di dalam array products
- Isi semua field: name, category, city, address, products

Hint:

```

db.toko.insertOne({
  "name"      : "...",
  "category"  : "...",
  "city"      : "...",
  "address"   : { "street" : "...", "zipcode" : "..." },
  "products" : [
    { "item" : "...", "price" : ..., "stock" : ... },
    { "item" : "...", "price" : ..., "stock" : ... }
  ]
})

```

```

> db.toko.insertOne({
  "name": "Toko Sentosa",
  "category": "Elektronik",
  "city": "Medan",
  "address": { "street": "Jl. Gajah Mada", "zipcode": "20111" },
  "products": [
    { "item": "Speaker Bluetooth", "price": 850000, "stock": 15 },
    { "item": "Powerbank 20k", "price": 350000, "stock": 25 }
  ]
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a1107e5f41c09c9d87537e6')
}
test>

```

2.) Tampilkan semua toko yang berada di kota Jakarta.

```
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```

```
> db.toko.find({ "city": "Jakarta" }).pretty()
```

```
< {
  _id: ObjectId('6a1107d1f41c09c9d87537e2'),
  name: 'Toko Berkah',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'TV',
      price: 3500000,
      stock: 10
    },
    {
      item: 'Kulkas',
      price: 2800000,
      stock: 5
    }
  ]
}
{
  _id: ObjectId('6a1107d1f41c09c9d87537e4'),
  name: 'Toko Salim Bakery',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
```

```

    {
      _id: ObjectId('6a1107d1f41c09c9d87537e4'),
      name: 'Toko Salim Bakery',
      category: 'Elektronik',
      city: 'Jakarta',
      address: {
        street: 'Jl. Sudirman',
        zipcode: '10220'
      },
      products: [
        {
          item: 'Monitor LED',
          price: 7700000,
          stock: 11
        },
        {
          item: 'Airfryer Panasonic',
          price: 4850000,
          stock: 7
        }
      ]
    }
  ]
}
test> |

```

3.) Tampilkan toko yang memiliki produk dengan price lebih dari 3.000.000.

Hint:

- Gunakan dot notation: products.price
- Gunakan operator: \$gt

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```
> db.toko.find({ "products.price": { "$gt": 3000000 } }).pretty()
```

```
< {
  _id: ObjectId('6a1107d1f41c09c9d87537e2'),
  name: 'Toko Berkah',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'TV',
      price: 3500000,
      stock: 10
    },
    {
      item: 'Kulkas',
      price: 2800000,
      stock: 5
    }
  ]
}
{
  _id: ObjectId('6a1107d1f41c09c9d87537e3'),
  name: 'Toko Maju',
  category: 'Elektronik',
  city: 'Bandung',
  address: {
```

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```
  _id: ObjectId('6a1107d1f41c09c9d87537e3'),
  name: 'Toko Maju',
  category: 'Elektronik',
  city: 'Bandung',
  address: {
    street: 'Jl. Asia Afrika',
    zipcode: '40111'
  },
  products: [
    {
      item: 'HP',
      price: 15000000,
      stock: 20
    },
    {
      item: 'Laptop',
      price: 80000000,
      stock: 3
    }
  ]
}
{
  _id: ObjectId('6a1107d1f41c09c9d87537e4'),
  name: 'Toko Salim Bakery',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
```

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```
  _id: ObjectId('6a1107d1f41c09c9d87537e4'),
  name: 'Toko Salim Bakery',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'Monitor LED',
      price: 7700000,
      stock: 11
    },
    {
      item: 'Airfryer Panasonic',
      price: 4850000,
      stock: 7
    }
  ]
}
{
  _id: ObjectId('6a1107d1f41c09c9d87537e5'),
  name: 'Toko Sudinotech',
  category: 'Elektronik',
  city: 'Surabaya',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
```

```

    _id: ObjectId('6a1107d1f41c09c9d87537e5'),
    name: 'Toko Sudinotech',
    category: 'Elektronik',
    city: 'Surabaya',
    address: {
      street: 'Jl. Sudirman',
      zipcode: '10220'
    },
    products: [
      {
        item: 'RTX 55000',
        price: 8500000,
        stock: 20
      },
      {
        item: 'Set PC Gaming Radeon',
        price: 12000000,
        stock: 9
      }
    ]
  }
}
test>

```

4.) Ubah category Toko Berkah dari Fashion menjadi Aksesoris.

Tambahkan 1 produk baru ke dalam array products milik Toko Maju menggunakan \$push.

Hint:

- Filter berdasarkan name: Toko Maju
- Gunakan operator \$push untuk menambah elemen ke array

```

> db.toko.updateOne(
  { "name": "Toko Berkah" },
  { "$set": { "category": "Aksesoris" } }
);

db.toko.updateOne(
  { "name": "Toko Maju" },
  { "$push": { "products": { "item": "Mouse Gaming", "price": 450000, "stock": 15 } } }
);
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test>

```

5.) Hitung jumlah toko per city, urutkan dari kota yang paling banyak tokonya.

Gunakan:

- \$group dengan \$sum: 1
- \$sort descending (-1)

```

> db.toko.aggregate([
  { "$group": { "_id": "$city", "total_toko": { "$sum": 1 } } },
  { "$sort": { "total_toko": -1 } }
])
< {
  _id: 'Jakarta',
  total_toko: 2
}
{
  _id: 'Medan',
  total_toko: 1
}
{
  _id: 'Surabaya',
  total_toko: 1
}
{
  _id: 'Bandung',
  total_toko: 1
}
test>

```

6.) Hitung rata-rata harga produk per category.

Tahapan pipeline:

- \$unwind: '\$products' (bongkar array products dulu)
- \$group: kelompokkan per category, hitung \$avg dari products.price
- \$sort: urutkan sesuai kebutuhan

```

> db.toko.aggregate([
  { "$unwind": "$products" },
  { "$group": { "_id": "$category", "rata_rata_harga": { "$avg": "$products.price" } } }
])
< {
  _id: 'Elektronik',
  rata_rata_harga: 4911111.111111111
}
{
  _id: 'Aksesoris',
  rata_rata_harga: 3150000
}

```

7.) Tambahkan 1 produk baru ke dalam array products milik Toko Maju menggunakan \$push.

Hint:

- Filter berdasarkan name: Toko Maju
- Gunakan operator \$push untuk menambah elemen ke array

```

> db.toko.updateOne(
  { "name": "Toko Maju" },
  { "$push": { "products": { "item": "Keyboard Mechanical", "price": 750000, "stock": 10 } } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test>

```

8.) Tampilkan harga produk tertinggi (\$max) per city.

Hint:

- Unwind products terlebih dahulu
- Gunakan \$max pada products.price

```

> db.toko.aggregate([
  { "$unwind": "$products" },
  { "$group": { "_id": "$city", "harga_tertinggi": { "$max": "$products.price" } } }
])
< {
  _id: 'Medan',
  harga_tertinggi: 850000
}
{
  _id: 'Bandung',
  harga_tertinggi: 8000000
}
{
  _id: 'Jakarta',
  harga_tertinggi: 7700000
}
{
  _id: 'Surabaya',
  harga_tertinggi: 12000000
}
test>

```

9.) Buat pipeline lengkap dengan 4 tahap:

- \$match — filter hanya toko category Elektronik
- \$unwind — bongkar array products
- \$group — hitung total stock per city
- \$sort — urutkan dari total stock terbesar

```

> db.toko.aggregate([
  { "$match": { "category": "Elektronik" } },
  { "$unwind": "$products" },
  { "$group": { "_id": "$city", "total_stok": { "$sum": "$products.stock" } } },
  { "$sort": { "total_stok": -1 } }
])
< {
  _id: 'Bandung',
  total_stok: 48
}
{
  _id: 'Medan',
  total_stok: 40
}
{
  _id: 'Surabaya',
  total_stok: 29
}
{
  _id: 'Jakarta',
  total_stok: 18
}
test>

```

10.) Hapus toko yang memenuhi dua kondisi sekaligus:

- category adalah Fashion
- city adalah Surabaya

Hint:

- Gunakan \$and atau tulis kedua kondisi dalam satu objek filter

```
> db.toko.deleteMany({
  "$and": [
    { "category": "Fashion" },
    { "city": "Surabaya" }
  ]
})
< {
  acknowledged: true,
  deletedCount: 0
}
test>
```